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Superpermeability to fast and thermal hydrogen particles: applications to the pumping and recycling of hydrogen isotopes

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Metal membranes (e.g. of Nb, Pd, Fe, etc.) may be, under certain conditions, superpermeable to hydrogen fast particles and thermal atoms, i.e. their permeability may approach that of an opening of the same area. A proposal is put forward that superpermeable membranes be installed in the ITER pumping ducts in order to isolate the major part of D/T mixture and thus to avoid its accumulation in the cryopumps that pump helium. Nonthermalized particles coming from the divertor region will be immediately pumped by the membrane, while thermalized H_2 molecules will be atomized at incandescent surfaces specially introduced. The latter mechanism is experimentally demonstrated on a Nb membrane, with hydrogen atomizers made of Nb and Ta. The specific pump speed reached amounts up to $\simeq 1.5\ l s^{-1}$ per 1 cm^2 of the membrane area. Pumping of hydrogen is accompanied by its automatic compression beyond the membrane by a factor of $\simeq 10^4$.

1. Superpermeability: physical idea

Metal membranes of usual thickness may become, under certain conditions, *superpermeable* (a term first suggested by one of the authors [1]) to hydrogen particles, i.e. their permeability may approach the conceivable limit: permeability of an opening of the same area. The physical ideas to explain this phenomenon can be briefly summarized as follows:

A potential barrier may be built up at a metal surface, originating from a nonmetallic impurity layer, that effectively impedes the dissociative uptake of molecular hydrogen by the metal. Such a barrier suppresses still further the metal permeability to hydrogen, small as it is by itself. *Suprathermal particles* (including *thermal H atoms*) are able to freely pass the barrier. The nonmetallic layer impedes, however, the back release of absorbed hydrogen particles through inhibiting their thermal recombination at the surface since the retained particles "cool down" quickly inside the metal lattice. The impediment to release at the membrane inlet surface can be made so great that the probability for a retained H atom to reach the opposite surface by interstitial diffusion becomes larger than the one to be desorbed into the gas at the inlet side. If the desorption at the outlet surface is easier (e.g. the barrier is lower there), then virtually all the particles retained will be released at the outlet side. In this way, the membrane becomes virtually "transparent" to incident suprathermal particles due to the "brightening" effect of a nonmetallic layer at the inlet surface.

With only thermal H_2 molecules being released at the outlet side, an automatic compression becomes possible of the permeating hydrogen due to a very small permeability of the membrane to molecular hydrogen gas. The degree of compression may amount to several orders of magnitude.

A comprehensive description of the above phenomenological concept is given in these authors' previous contributions (e.g. refs. [2–4]). Numerous experimental data on superpermeability have been obtained by the authors [1,4–10] and by Waelbroeck et al. [11,12]. A better understanding of surface chemistry effects on retention, release and permeation of suprathermal hydrogen is also due to the works of Pick and Sonnenberg [13], Yamawaki et al. [14], Causey et al. [15], Kerst and Swansiger [16] and Wampler [17].

2. Superpermeability: experimental evidence

New data on ion-beam experiments obtained by the authors with the use of an experimental apparatus described earlier [10] are presented in figs. 1 and 2. In ultrahigh vacuum, switching on the 19.2 keV H_3^+ ion beam (6.4 keV per proton) acting upon a 100 μm thick Nb membrane results (fig. 1) in a rapid establishment of the re-emission flux that virtually equals the incident

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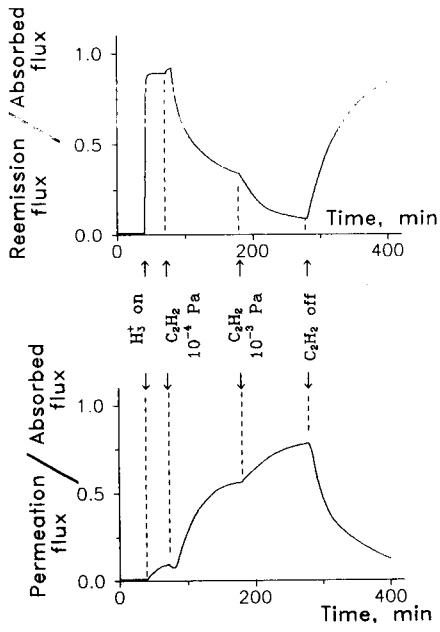


Fig. 1. Effects of switching of the ion beam and of admission of acetylene on the re-emission and permeation of hydrogen in experiments with a 100 μm thick Nb membrane under the action of a 19.2 keV H_3^+ ion beam (2.5×10^{15} $Hcm^{-2}s^{-1}$). The membrane temperature is 893 K.

flux, whereas only a very small portion of the flux passes through the membrane. Admission of acetylene at the inlet side leads to suppression of re-emission,

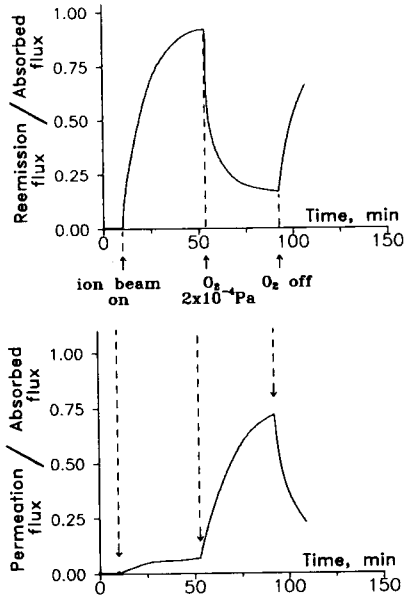


Fig. 2. Effects of switching of the ion beam and of admission of oxygen on the re-emission and permeation of hydrogen in experiments with a 100 μm thick Nb membrane under the action of a 200 eV H_3^+ ion beam (2×10^{15} $Hcm^{-2}s^{-1}$). The membrane temperature is 838 K.

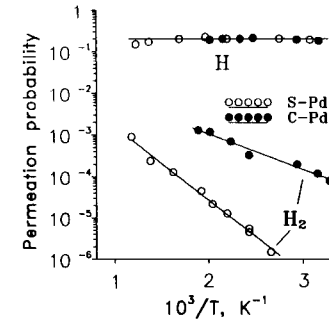


Fig. 3. Arrhenius plots for the probabilities of permeation, per single impact on the membrane surface, for H atoms and H_2 molecules acting on Pd membranes covered by S [8] and C [9] monolayers.

and to a dramatic increase of the membrane's permeability up to its almost total "transparency" due to the deposition of a carbon film by acetylene pyrolysis.

Fig. 2 illustrates a similar result of the experiment with a low energy (200 eV H_3^+, 67 eV per proton) ion beam, and with oxygen instead of carbon.

The authors' experimental data pertaining to Pd membranes covered by monolayers of S [8] and C [9] are shown in fig. 3. The probability of absorption of a H atom, per single impact on the membrane surface ($\simeq 0.2$), does not depend on the metal temperature (over the range stretching down to room temperature), and is insensitive to the chemical composition of the surface. Virtually every atom absorbed penetrates through the membrane. The membrane permeability to H atoms exceeds its permeability to molecular hydrogen gas by several orders of magnitude. A similar result is obtained with a Fe membrane [6] that has an not completely cleaned surface (fig. 4). The difference in permeability to atoms and molecules reaches up to 10^{10} in this case, which would permit a 10^{10} degree of compression. It is important to note that the nonmetallic monolayers responsible for superpermeability are absolutely stable under the action of thermal atoms

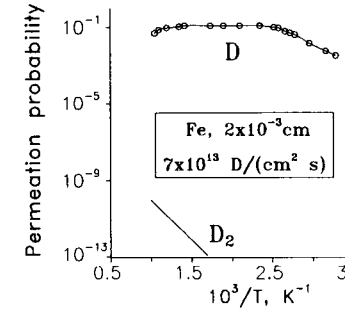


Fig. 4. Arrhenius plots for the probabilities of permeation, per single impact on the membrane surface, for H atoms and H_2 molecules acting on a Fe membrane covered by residual impurities [6].

over the whole ranges of metal temperature and atomic flux investigated.

3. Application of superpermeability to the ITER project

The superpermeability phenomenon can be used in fusion devices to evacuate, isolate and recycle a D/T mixture (e.g. at several points in the beam injectors, in mirror devices, etc.). A suggestion [18,19] to employ superpermeable membranes for the ITER exhaust pumping has been proposed [#1].

Cryogenic pumps are presently chosen as the most suitable means to pump helium ashes at the ITER exhaust. One disadvantage of this option is that substantial amounts of tritium will be accumulated at the cryopanels, which is both very expensive and leads to the danger of radioactive spillage at accidental unfreezing of the panels. Separation of the major part of the D/T mixture by membranes located before the cryopumps may effectively decrease the tritium accumulation.

It is suggested that the membranes be installed along the walls of every pumping duct, as shown in fig. 5. The membranes will isolate the major part of the D/T mixture out of the divertor exhaust. The flow of D/T, separated and compressed by the membranes, can then be directly returned to the fueling systems. The nonthermalized particles reaching the membrane in zone I (nearest to the divertor) after only a few bounces upon the duct wall will be immediately pumped due to their remaining excess of energy. Thermalized H_2 molecules will be atomized at the incandescent surfaces introduced in the duct in zone II, and H atoms will be pumped by the membrane mounted there.

[#1] It has also been suggested earlier by Sugihara and Abe [20] to employ metal membranes for D/T mixture separation from He and for its return from the divertor chamber directly to the working plasma of a tokamak.

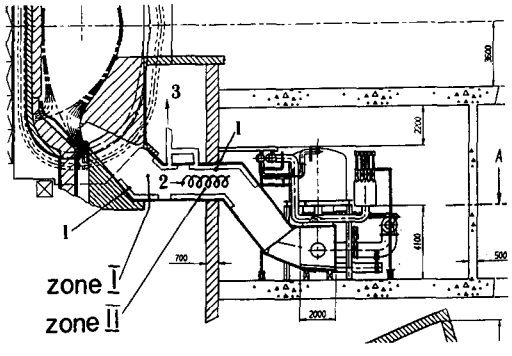


Fig. 5. Suggested location of superpermeable membranes (1) and of atomizers (2) in the ITER pumping ducts. The membranes in zone I (out of line of sight with the divertor plate) pump suprathermal particles coming from the divertor region, while the membranes in zone II pump H atoms produced by the atomizers. The D/T mixture separated and compressed by the membranes is then returned to the fuel cycle through the output tubes (3).

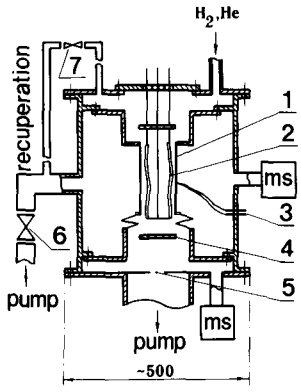


Fig. 6. The UHV apparatus for model experiments. (1) Nb membrane, 10 cm diam., 20 cm long, 100 μm thick; (2) atomizers (four resistively heated ribbons of Nb, Ta and W foil); (3) thermocouple; (4) quartz shields (to diminish the losses by recombination); (5) calibrated diaphragm, with a conductance of 100 ls^{-1} for H_2; (6) valve to control the downstream chamber pumping; (7) valve to demonstrate the possibility of fuel recurperation; (ms) mass-spectrometric gas analyzers.

The atomizer should be placed directly facing the membrane [12] (otherwise, the losses by recombination at the walls will be unacceptably high); hence the difficulty arises from the deposition of atomizer material on the membrane surface. For example, evaporation of W might have disastrous effects on superpermeability. Fortunately, the V group metals (V, Nb and Ta) are among the most suitable candidates to use for superpermeable membranes; the two latter being refractory metals, they can be also employed as atomizers operating at $T \cong 1900$ K.

4. Experimental apparatus for model experiments [19]

An UHV vacuum vessel, in the form of a 50 cm diam. tube (fig. 6), models a stretch of the ITER pumping duct. A cylindrical membrane may be mounted just along the vessel walls, but we are presently working with a 10 cm diam., 20 cm long membrane made of 100 μm Nb foil. Four ribbons of Nb, Ta and W serve as atomizers.

The effects of switching on the atomizers are shown in fig. 7. The inlet pressure decreases, and the outlet pressure correspondingly increases to reach a steady state level, due to permeation of hydrogen through the membrane. Remarkably, the inlet pressure does not

rise back after the initial drop, which demonstrates that the re-emission flux is virtually zero. That means, in other words, that almost the whole flux absorbed penetrates through the membrane.

The pumping speed by the membrane is proportional to the atomizer surface area (this was checked by switching the atomizer ribbons one by one). The pump speed amounts to 800–1000 $l s^{-1}$ for a ratio between the atomizer and the membrane area of 0.15, which corresponds to about 1.5 $l s^{-1}$ per cm^2 of the membrane, or to about 10 $l s^{-1}$ per cm^2 of the atomizer.

The degree of compression of $\simeq 10^4$ was reached with no substantial decrease in the pumping speed (see fig. 7).

Helium, as one would expect, has no effect whatever on hydrogen pumping. It was tested in experiments with admission to the inlet side of helium at pressures twice as large as the hydrogen pressure.

The probabilities of an H_2 molecule atomization on Ta and Nb have been measured in these experiments (fig. 8). Their values are high enough, and are close to the known values for tungsten.

One important conclusion follows from the analysis of the above experimental data. The H atoms that are not absorbed at the first impact on the membrane do not recombine, and may be absorbed at repeated collisions. If this were not true, a much larger atomizer area would be required to provide for the recurrent atomization, which might demand, in turn, an intolerably high consumption of power for the atomizer heating.

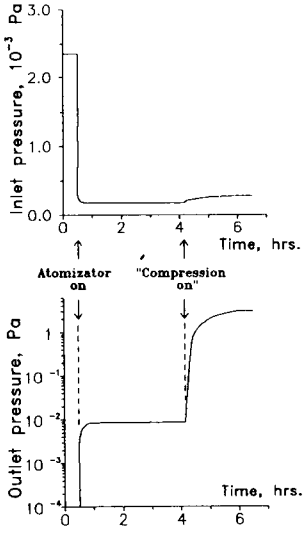


Fig. 7. Effect of switching on the atomizer on hydrogen pressure in the upstream and downstream chambers. The "compression on" arrow indicates the moment when the downstream chamber valve ((6) in fig. 6) is partly closed.

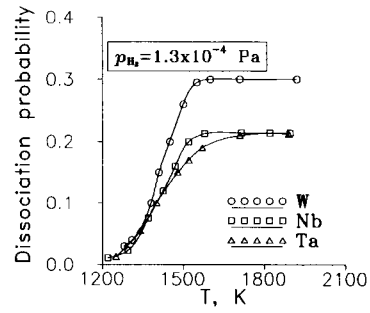


Fig. 8. Temperature dependence of the probability of H_2 molecule dissociation on Nb and Ta compared to that on W.

5. Discussion

Listed below are some operational parameters expected for the D/T mixture pumping system (fig. 5) suggested for ITER:

a) Specific pumping speed per 1 m^2 of the membrane: 12–14 $m^3 s^{-1}$.

b) Power to be consumed per unit pumping speed: 1.7×10^{-3} MW/($m^3 s^{-1}$).

To isolate 75% of D/T mixture by the membrane (with no account for nonthermalized particles pumped in zone I) one requires: a full membrane area of 160 m^2; a membrane length (in each duct) of 1.6 m; and a full power consumed of 5 MW. A degree of compression of 10^4–10^5 may be achieved under such conditions; an accumulation of tritium in a 100 μm thick membrane will amount to 5–10 g at the 10^3 degree of compression.

If at least half of the particles from the divertor region were reaching zone I (fig. 5) without yet being thermalized and were pumped there, the portion of D/T mixture evacuated by the membrane might approach 90%.

6. Conclusion

The parameters indicated above appear rather promising. We would deem it reasonable, therefore, to start working on the ITER pumping system design and manufacture technology, along with proceeding with the model experiments.

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